FBS Physiology — NBME Study Guide

Homeostasis · Body fluids & edema · Membrane transport · Resting membrane potential · Action potentials & conduction

Red = extra high yield (classic NBME) Black = core content *Italic blue = mnemonic*

1. Homeostasis & control systems

- **Homeostasis** = keeping the **internal environment (ECF)** constant (Claude Bernard's "milieu intérieur").
- **Negative feedback** = response *opposes* the change → stability. Most body systems (BP via baroreceptors, temperature, blood glucose, PTH–Ca^{2+}, ADH–osmolarity).
- **Positive feedback** = response *amplifies* the change. Physiologic examples: **LH surge (estrogen), oxytocin in labor (Ferguson reflex), clotting cascade, action potential upstroke (Na^+ channel opening), lactation**. Pathologic: "vicious cycle" in hemorrhagic shock (cardiac failure → ↓ perfusion → worse failure).
- **Feedforward** = anticipatory response before the variable changes (e.g., ↑ HR before exercise, cephalic phase of digestion).
- **Gain = correction ÷ remaining error**. Higher gain = better control. Example: BP pushed +100, baroreflex returns to +25 → correction 75 / error 25 → gain = −3 (negative = negative feedback). Temperature control has very high gain (~−33).
- **Baroreceptor reflex**: carotid sinus (CN IX) & aortic arch (CN X) stretch → ↑ BP ⇒ ↑ firing ⇒ ↑ vagal, ↓ sympathetic ⇒ ↓ HR, vasodilation. **Fast, short-term BP control; resets in chronic HTN.** Carotid massage ↑ firing ⇒ ↓ HR.
- Chemoreceptors (carotid/aortic bodies) sense ↓ PO_2, ↑ PCO_2, ↓ pH.
- Kidneys = long-term BP control (pressure natriuresis, RAAS); infinite gain over time.

2. Body fluid compartments

Compartment	% body weight	Fraction	Measured with
Total body water (TBW)	**≈ 60% (60–40–20 rule)**	—	D_2O, tritiated water, antipyrine
ICF	40%	**2/3 of TBW**	TBW − ECF
ECF	20%	**1/3 of TBW**	Inulin, mannitol, ^{22}Na
Interstitial fluid	15%	**3/4 of ECF**	ECF − plasma
Plasma	5%	**1/4 of ECF**	Evans blue, ^{125}I-albumin

- Blood volume = plasma volume ÷ (1 − Hct). TBW is **lower in women, obese, elderly** (fat has little water); higher in infants (~75%).
- **Main ions: ECF = Na^+, Cl^-, HCO_3^-. ICF = K^+, Mg^{2+}, phosphates, proteins.** Plasma has more protein than interstitial fluid (Gibbs–Donnan effect: slightly more cations in plasma).
- **Osmolarity** (osm/L solution) vs **osmolality** (osm/kg water); normal ≈ **285–295 mOsm/kg**. Plasma Osm ≈ $2[Na^+]$ + glucose/18 + BUN/2.8.
- **Tonicity** = effective osmolarity (only *non-penetrating* solutes count). **Urea is an ineffective osmole** (crosses membranes) → isosmotic urea solution is *hypotonic* → cells swell.
- Cell in **hypotonic** solution → water in → swells/lyses. **Hypertonic** → water out → shrinks (crenation). 0.9% NaCl & 5% dextrose (initially) are isotonic.
- **Water always moves toward higher osmolarity; disturbances start in the ECF, and the ICF changes only secondarily by osmosis.**

Darrow–Yannet diagrams (volume/osmolarity shifts) — MUST KNOW

Type	Example	ECF vol	ICF vol	Osmolarity	Hct / plasma protein
Isosmotic expansion	Isotonic saline infusion	↑	—	—	↓ / ↓
Isosmotic contraction	**Diarrhea, burns, vomiting**	**↓**	**—**	**—**	**↑ / ↑**
Hyperosmotic expansion	High NaCl intake, hypertonic saline	↑	↓	↑	↓ / ↓

Hyperosmotic contraction	**Sweating, fever, diabetes insipidus**	**↓**	**↓**	**↑**	**— / ↑**
Hyposmotic expansion	**SIADH, water intoxication**	**↑**	**↑**	**↓**	**— / ↓**
Hyposmotic contraction	**Adrenal insufficiency (↓ aldosterone)**	**↓**	**↑**	**↓**	**↑ / ↑**

Hct logic: Hct = RBC volume ÷ (RBC + plasma). Losing plasma alone (diarrhea) ↑ Hct. RBCs act like ICF: they shrink when ECF osmolarity ↑ and swell when it ↓, which cancels or adds to the plasma change. Acute hemorrhage (whole blood lost) → ECF ↓ but Hct unchanged at first.

- Heavy sweating → **ADH ↑**, urine output drops (≈1,500 → 500 mL/day), thirst ↑.
- **Hyponatremia** = too much water relative to Na^+ (dilutional), e.g., SIADH → brain swelling (confusion, seizures). Correct slowly (osmotic demyelination).

Starling forces & edema

- **Net filtration = Kf [(Pc − Pi) − σ(πc − πi)]**. Pc pushes out; πc (plasma oncotic, mainly **albumin**) pulls in.
- **Edema causes: ↑ Pc (heart failure, venous obstruction, DVT), ↓ πc (nephrotic syndrome, liver cirrhosis, malnutrition/kwashiorkor), ↑ Kf/permeability (burns, inflammation, sepsis), lymphatic obstruction (filariasis, cancer, node dissection → non-pitting).**
- Safety factors against edema: low interstitial compliance, ↑ lymph flow (up to 10–50×), washout of interstitial protein.
- Lymph returns ~2–4 L/day of filtered fluid and protein; thoracic duct → left subclavian/IJ junction.

3. Membrane transport

Type	Energy	Carrier?	Saturable?	Examples
Simple diffusion	None (down gradient)	No	No (linear)	O_2, CO_2, steroids, ethanol, urea (some)
Facilitated diffusion	None	Yes	**Yes (Tm)**	**GLUT1–5 (GLUT4 insulin-dependent: muscle, fat)**
Channels	None	Pore	Can saturate at very high conc.	Na^+/K^+/Ca^{2+} channels, aquaporins
Primary active	**ATP directly**	**Yes**	**Yes**	**Na^+/K^+-ATPase, Ca^{2+}-ATPase (SERCA, PMCA), H^+/K^+-ATPase**
Secondary active	**Na^+ gradient (ATP indirectly)**	**Yes**	**Yes**	**SGLT (symport), Na^+/Ca^{2+} exchanger NCX & Na^+/H^+ exchanger NHE (antiport)**

- **Fick's law**: diffusion rate ∝ (area × concentration gradient × permeability)/thickness. Lipid solubility ↑ permeability; size and charge ↓ it. **Ions can't cross the bilayer because they are charged/hydrated, not lipid-soluble.**
- **Na^+/K^+-ATPase: 3 Na^+ OUT, 2 K^+ IN per ATP** *(NOKIA: Na^+ Out, K^+ In, ATPase)*. Basolateral in epithelia. Electrogenic (net +1 out) → contributes a few mV to RMP. Inhibited by **digoxin/ouabain**.
- **Symporter** = same direction (SGLT: Na^+ + glucose IN). **Antiporter** = opposite directions (NCX: 3 Na^+ in, 1 Ca^{2+} out; NHE: Na^+ in, H^+ out).
- **Proximal tubule glucose reabsorption order: (1) basolateral Na^+/K^+-ATPase pumps Na^+ out → (2) low intracellular Na^+ creates gradient → (3) apical SGLT2 drags glucose in against its gradient → (4) GLUT2 lets glucose out basolaterally.**
- Glucose transport saturates: renal threshold ≈ **200 mg/dL**, Tm ≈ 375 mg/min → glucosuria in diabetes.
- **SGLT2 inhibitors (-gliflozins)** → glucosuria; side effects UTIs, genital yeast infection, euglycemic DKA, volume depletion.
- **Blocking the Na^+/K^+ pump (or ischemia/no ATP) → pump fails FIRST, secondary transport (SGLT) fails second**, cells swell (Na^+ + water in).

Digitalis (digoxin) cascade — classic NBME: blocks Na^+/K^+-ATPase → ↑ intracellular Na^+ → weaker Na^+ gradient → NCX removes less Ca^{2+} → ↑ intracellular Ca^{2+} → more SR Ca^{2+} → **↑ contractility (positive inotropy)**. Also ↑ vagal tone (slows AV node). **Hypokalemia potentiates toxicity** (K^+ competes for the same site). Toxicity: nausea, yellow-green vision, arrhythmias; antidote = digoxin Fab.

- **Osmosis**: water through aquaporins toward higher osmolarity. Osmotic pressure π = g·C·σ·RT. Reflection coefficient σ: 1 = impermeant (albumin), 0 = freely permeant (urea).
- **Vesicular transport**: endocytosis (phagocytosis, pinocytosis, receptor-mediated via clathrin — LDL), exocytosis (Ca^{2+}-triggered, SNAREs), transcytosis.

4. Resting membrane potential (RMP)

- **At rest the membrane is most permeable to K^+ (K^+ leak channels) → RMP ≈ −70 to −90 mV, close to E_K.** The Na^+/K^+ pump *maintains* gradients (and adds ~−4 mV) but is not what directly sets RMP.
- **Nernst equation** (single ion at 37 °C): E = (61.5/z) log([out]/[in]).
- **Goldman (GHK) equation**: RMP = weighted average of E for each ion by its permeability → moves toward the ion with highest permeability.

Ion	[ECF]	[ICF]	Equilibrium potential (deck values)
K^+	**4–5 mM**	**140–150 mM**	**≈ −95 mV**
Na^+	**140–145 mM**	**10–15 mM**	**≈ +65 mV**
Cl^-	~105–110 mM	~5–10 mM	≈ −90 mV
Ca^{2+}	~2.5 mM	~0.0001 mM	≈ +120 mV

- **Hyperkalemia → smaller K^+ gradient → less K^+ efflux → depolarized RMP (closer to threshold)**. Initially excitable, but sustained depolarization inactivates Na^+ channels → weakness, **peaked T waves, wide QRS, arrhythmias**. Treat: **IV calcium gluconate** (stabilizes membrane), insulin + glucose, β_2-agonists (shift K^+ in), binders, dialysis.
- **Hypokalemia → hyperpolarized RMP (further from threshold)** → muscle weakness, cramps, ileus, flat T waves, **U waves**.
- **Calcium sets the THRESHOLD (stabilizes Na^+ channels).** **Hypocalcemia** → Na^+ channels open with less depolarization → threshold drops toward RMP → spontaneous firing → **tetany, paresthesias, Chvostek & Trousseau signs** (e.g., after thyroidectomy/hypoparathyroidism). **Hypercalcemia** → threshold ↑ → ↓ excitability, **sluggish reflexes**, weakness *(stones, bones, groans, thrones, psychiatric overtones)*.

5. Action potentials & conduction

Phases (nerve / skeletal muscle)

- **Threshold** (~−55 mV) → voltage-gated Na^+ channel **activation (m) gates** open (positive feedback) → **upstroke = Na^+ influx**; overshoot to ~+35 mV.
- At peak, **Na^+ inactivation (h) gates close** → Na^+ entry stops; they stay closed until repolarization → **absolute refractory period**.
- **Repolarization = K^+ efflux** through delayed-rectifier voltage-gated K^+ channels; slow closure → **after-hyperpolarization** (undershoot toward E_K).
- **Relative refractory period**: some Na^+ channels recovered; a stronger-than-normal stimulus can fire.
- **All-or-none**; stimulus intensity is coded by **frequency** of APs, not amplitude.
- **Accommodation**: slow depolarization (e.g., hyperkalemia) inactivates Na^+ channels before threshold → no AP even though Vm passes threshold.

Toxin / drug	Target	Effect
Tetrodotoxin (pufferfish), saxitoxin (red tide)	**Voltage-gated Na^+ channel (from outside)**	**No upstroke → no AP, paralysis**
Tetraethylammonium (TEA)	**Voltage-gated K^+ channels (from inside)**	**Impaired repolarization → prolonged AP**
Local anesthetics (lidocaine)	Na^+ channel (from inside, use-dependent)	Block conduction; small, myelinated pain fibers first
Ciguatoxin / batrachotoxin	Hold Na^+ channels open	Persistent depolarization

Conduction velocity

- **↑ velocity with ↑ diameter (↓ internal resistance) and myelination.**

- **Myelin ↓ membrane capacitance and ↑ membrane resistance** → less charge needed to change Vm, current spreads farther → **saltatory conduction** node to node (Na^+ channels concentrated at nodes of Ranvier).
- Length constant $\lambda = \sqrt{(Rm/Ri)}$: large λ = signal spreads farther. Time constant $\tau = Rm \times Cm$: small τ = faster depolarization.
- **Demyelination (MS — CNS oligodendrocytes; Guillain–Barré — PNS Schwann cells) ↑ capacitance, current leaks → slowed/blocked conduction.**
- Fiber types: Aα (largest, fastest: proprioception, motor) > Aβ (touch) > Aγ > Aδ (fast pain, cold) > B (preganglionic autonomic) > C (unmyelinated, slow pain, postganglionic).

Graded vs action potentials

Graded (local) potential	Action potential
Size ∝ stimulus; summate (spatial/temporal)	All-or-none
Decay with distance; no refractory period	Propagated without decrement; refractory period
EPSP/IPSP, receptor/end-plate potentials	Axon, muscle fibers

6. Rapid-fire NBME pearls

- **Na^+/K^+ pump: 3 out, 2 in, primary active, basolateral; digoxin & ouabain block it.**
- **SGLT = secondary active symport (apical); GLUT = facilitated diffusion, no ATP.**
- **RMP ≈ E_K because K^+ permeability dominates; ECF K^+ is the main determinant of RMP stability.**
- **TTX blocks Na^+ (no upstroke); TEA blocks K^+ (no repolarization).**
- **Diarrhea = isosmotic contraction (↑ Hct). SIADH = hyposmotic expansion. Adrenal insufficiency = hyposmotic contraction. Sweating = hyperosmotic contraction.**
- **Nephrotic syndrome / cirrhosis → ↓ albumin → ↓ πc → edema. Heart failure → ↑ Pc → edema.**
- ICF 2/3, ECF 1/3; interstitial 3/4 of ECF, plasma 1/4.
- Positive feedback examples: LH surge, oxytocin, clotting, Na^+ channel opening in AP upstroke.
- Hypocalcemia → tetany (↓ threshold). Hypercalcemia → sluggish reflexes.